Two-Slit Diffraction and Missing Orders

SYNOPSIS

Note 10 treated the slits as points and Note 11 treated one slit as wide. Real slits are both. The pattern you actually see is the two-slit interference pattern **multiplied by** the single-slit envelope — and where the envelope happens to be zero, a bright fringe that Note 10 promised you simply is not there.

BEFORE YOU START — PREDICT

You count the bright fringes on the screen and find … **the third one on each side is missing**. Orders 1, 2, 4, 5 are all present. Nothing is wrong with the apparatus. What could put a hole in an otherwise regular row of fringes?

PERFORMANCE EXPECTATIONS

1. **State** the two-slit pattern as a product of an interference term and a diffraction envelope.
2. **Derive** $\beta = m\pi D/d$ at the position of the mth interference maximum.
3. **Predict** which orders are missing from the ratio d/D.
4. **Count** how many interference fringes fall inside the central diffraction peak.

The two effects together

THE RULE

The diffraction pattern of two slits of width D separated by a distance d is the interference pattern of two point sources separated by d **multiplied by** the diffraction pattern of a slit of width D

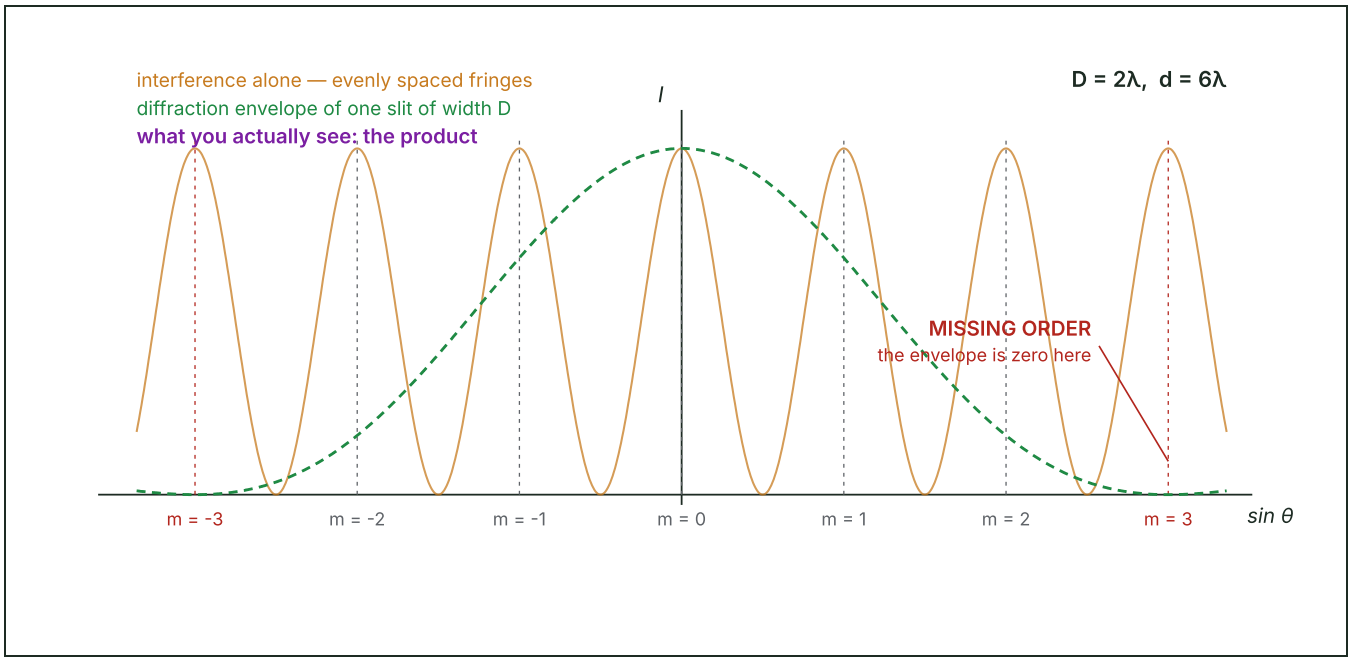


Figure 1. The two patterns, and their product. Every curve here is sampled from its own formula — **the missing order is missing because the arithmetic kills it**, not because it was left out of the drawing.

Your job on Figure 1. The interference fringes and the diffraction envelope are both drawn for you. Draw the curve you would **actually see** — multiply them, point by point. Which fringes disappear, and why those?

As D grows the envelope narrows — a wide slit diffracts less — so fewer and fewer fringes survive underneath it. Shrink D and the envelope opens out, letting more through.

Why an order goes missing

Take the two conditions and put them together. **Interference** gives a maximum at

$d \sin\theta = m\lambda$ so $\sin\theta = m\lambda/d$

and the **diffraction** intensity at that angle is $I = I_0(\sin\beta/\beta)^2$ with $\beta = \pi D \sin\theta/\lambda$. Substituting one into the other:

$$\beta = (\pi D/\lambda)(m\lambda/d)\ \beta = m\pi D / d$$

and the wavelength has **cancelled**. Whether an order survives depends only on the ratio of the slit width to the slit spacing.

1. Show that $m = 3$ is the missing order when $D = 2\lambda$ and $d = 6\lambda$, and find the intensity of the $m = 1$ fringe.

How many fringes fit inside the central peak?

Young's two-slit interference. Slit width D = **0.02 mm**, separated by d = **0.2 mm**, with λ = **500 nm**. How many fringes are observed in the central peak of the diffraction pattern?

> **HINT**
>
> Find the **first dark diffraction** — the edge of the central peak — and then ask how many interference orders fit inside that angle.

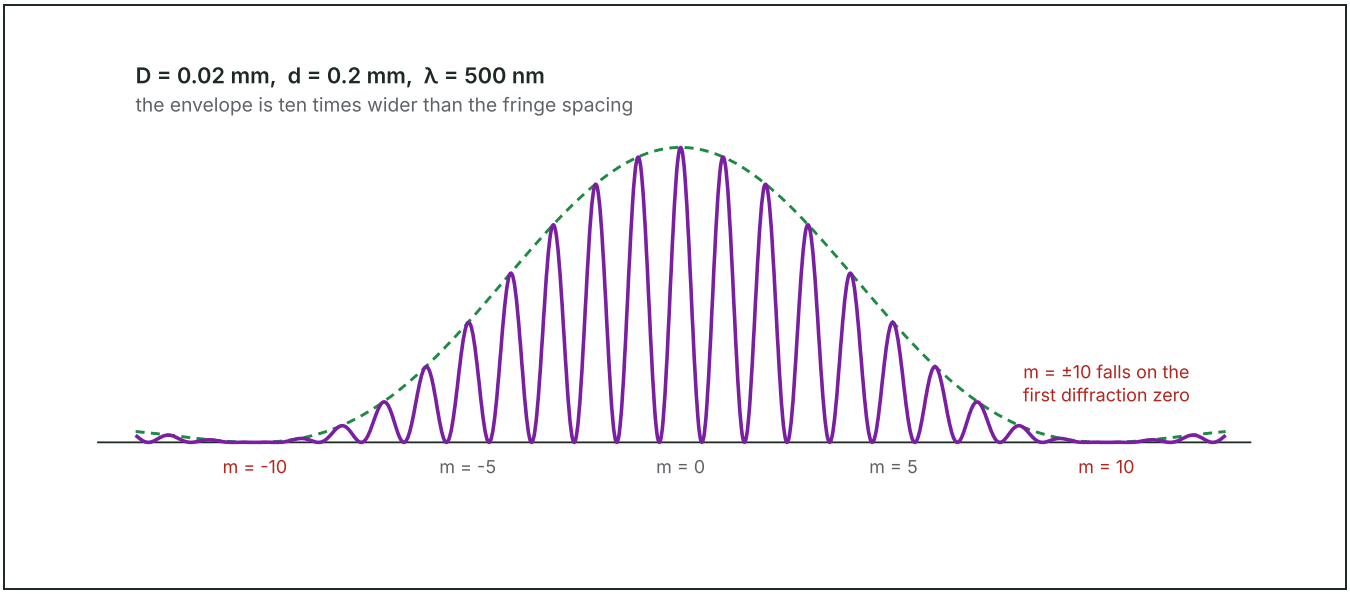


Figure 2. Narrow the slits relative to their spacing and the envelope opens out, letting many fringes through. **The ratio d/D = 10 is exactly the order that goes missing** — which is the worked example below, drawn.

THE PUNCHLINE

$$\beta = m\pi D/d \ \text{—} \ \text{the wavelength cancels}$$

Back to the prediction: a hole in a regular row of fringes means an **interference maximum has landed on a diffraction minimum**. The missing order is d/D, and it depends on the geometry of the slits alone — change the colour of the light and the same order is still missing.

Looking ahead

Two slits give fringes you have to look hard at. **Note 15** uses **ten thousand** of them per centimetre, which makes the maxima so sharp and so far apart that each wavelength in a beam separates out on its own.

2. Slits of width $D = 3\lambda$ are separated by $d = 12\lambda$. Which is the lowest missing order?

3. In $\beta = m\pi D/d$, what happened to the wavelength?

○ It was assumed to be 500 nm. ○ It cancelled, so which orders are missing depends only on d/D. ○ It is hidden inside β. ○ It only cancels for small angles.

4. For $D = 2\lambda$ and $d = 6\lambda$, what is the intensity of the $m = 1$ fringe, as a multiple of I_0?

Check Clear

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